ANSWER KEY & MARKING SCHEME SET A

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 20 N	F = ma
2	**(d)** 5 m s⁻²	a = F/m
3	**(a)** 50 N	W = mg
4	**(b)** 0 N	add if same direction; subtract if opposite
5	**(b)** 5 kg	m = F/a
6	**(c)** 1 kg	definition
7	**(b)** zero	Newton's first law
8	**(c)** increase the time of stopping and reduce the force	smaller acceleration
9	**(b)** upward on the rocket and greater than its weight	third law
10	**(b)** balanced	net force is zero

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	An object at rest remains at rest and an object in motion continues with constant velocity unless a net force acts on it (2).	2
12	The tendency of a body to resist a change in its state of rest or uniform motion (1); it depends on mass (1).	1+1
13	Inertia of motion: the body tends to continue its motion while the bus stops (2).	2
14	Acceleration is proportional to the net force and inversely proportional to mass, in the direction of the force (1); F = ma (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Push a stack of coins with a stretched rubber band on different surfaces (1); the distance travelled differs: larger on smoother surfaces (1); so friction is smaller on smooth surfaces (1).	1+1+1
16	A body on a smooth surface travels farther as friction decreases (1.5); in the limit of zero friction it would move forever with constant velocity, so no force is needed to maintain motion (1.5).	1½+1½
17	No (1); the applied force balances friction, so net force is zero (2).	1+2
	OR: At rest: horizontal line at one position, zero velocity (1.5); constant velocity: straight inclined line for position and horizontal line for velocity (1.5).	1½+1½
18	a = 50/10 = 5 m s⁻² (1.5); s = ½ × 5 × 16 = 40 m (1.5).	1½+1½
	OR: a = −20/5 = −4 m s⁻² (2); F = 1000 × (−4) = −4000 N, i.e. 4000 N opposite to motion (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A body continues in rest or uniform motion unless a net force acts (1). (ii) With less friction a body goes farther; with none it would move forever (2). (iii) Passenger lurching when the bus stops; dust shaken off a carpet (2).	1+2+2
	OR: (i) A cart pulled by a thread over a pulley with cup masses: doubling the load in the cup increases acceleration (2). (ii) Heavier cart, smaller acceleration (1). (iii) a = F/m or F = ma (2).	2+1+2
20	(i) Net force is zero, the block stays at rest (1). (ii) 5 N (1). (iii) 0.2 m s⁻² (1). (iv) ½ × 0.2 × 4 = 0.4 m (2).	1+1+1+2
	OR: (i) Every action has an equal and opposite reaction on the other body (1). (ii) Paddle pushes water back, water pushes paddle forward (2). (iii) They act on different objects (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 30 N	F = ma
2	**(d)** 5 m s⁻²	a = F/m
3	**(b)** 200 N	W = mg
4	**(a)** 17 N	add if same direction; subtract if opposite
5	**(b)** 12 m s⁻¹	a = F/m, v = at
6	**(c)** inertia	tendency to resist change
7	**(d)** Paddle pushes water backwards; water pushes paddle forwards	act on two different bodies
8	**(b)** increasing the time over which the passenger stops	reduces force
9	**(d)** 9.8 m s⁻²	acceleration due to gravitational force
10	**(b)** move in the opposite direction	third law

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The force that produces an acceleration of 1 m s⁻² in a body of mass 1 kg (2).	2
12	F = ma = 1500 × 2 (1) = 3000 N (1).	1+1
13	Whenever one object exerts a force on a second, the second exerts an equal and opposite force on the first (2).	2
14	They act on two different objects (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Net force = 5 N, a = 5/25 = 0.2 m s⁻² (2); s = ½ × 0.2 × 4 = 0.4 m (1).	2+1
16	The heavier child; F = ma, so for equal acceleration force is larger for larger mass (3).	3
17	Airbags and seat belts increase stopping time, reducing acceleration and force (1.5); a fielder pulling hands back while catching a cricket ball (1.5).	1½+1½
	OR: Weight = 30 × 9.8 = 294 N (2); she applies an equal upward force of 294 N (1).	2+1
18	The paddle pushes water backwards (1); water pushes the paddle forwards with an equal force (1); the forces act on different objects, so the canoe moves forward (1).	1+1+1
	OR: The foot pushes the ground backwards (1); friction from the ground acts forward on the foot (1); without friction we would slip (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) a = 2, F = 3000 N east (1). (ii) 0 (1). (iii) a = −2, F = −3000 N (westwards) (1). (iv) 25 + 50 + 25 = 100 m (2).	1+1+1+2
	OR: (i) Longer stopping time gives smaller acceleration (1). (ii) a = 300 m s⁻², F = 45 N (2). (iii) a = 3000, F = 450 N (2).	1+2+2
20	(i) Exhaust gases pushed down; the upward reaction exceeds weight (2). (ii) Engines fire against the direction of motion, slowing it (1). (iii) Air rushes out, balloon moves opposite (2).	2+1+2
	OR: (i) 2 N (1). (ii) 20 m s⁻² (1). (iii) 0.4 m s⁻² (1). (iv) Same force but different masses (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.